The Seven-Dimensional Case of the Skew-Hermitian Lie Algebra Problem

FA/Banach discovery run `fa_banach_001`

Candidate packet, 2026-06-13

Source problem

The source paper is V. Kadets, M. Martin, and R. Paya, "Recent progress and open questions on the numerical index of Banach spaces", arXiv:math/0605781. It asks for the possible dimensions of $\mathcal{Z}(X)$, the Lie algebra of skew-hermitian operators, when X is a finite-dimensional real Banach space.

> Our next aim is to discuss some questions related to the Lie algebra of skew-hermitian operators $\mathcal{Z}(X)$ of an arbitrary n-dimensional space which. The main related open question is the following.
>
> **Problem 27.** *Figure out what are the possible values for the dimension of $\mathcal{Z}(X)$ when $\dim(X) = n$.*
>
> Let us fix a n-dimensional Banach space X. It follows from a theorem of Auerbach [72, Theorem 9.5.1], that there exists an inner product $(\cdot|\cdot)$ on X such that every skew-hermitian operator on X remains skew-hermitian (hence skew-symmetric) on $H := (X, (\cdot|\cdot))$. Then, by just fixing an orthonormal basis of H, we get an identification of $\mathcal{Z}(X)$ with a Lie subalgebra of the Lie algebra $\mathcal{A}(n)$. Therefore,
>
> $$\dim\big(\mathcal{Z}(X)\big) \leqslant \frac{n(n-1)}{2}.$$
>
> The equality holds if and only if X is a Hilbert space (see [73, Theorem 3.2] or [60, Corollary 2.7]). It is a good question whether or not all the intermediate numbers are possible values for the dimension of $\mathcal{Z}(X)$. The answer is negative, as a consequence of Theorem 3.2 in Rosenthal's paper [73], which reads as follows.
>
> (a) If $\dim\big(Z(X)\big) > \frac{(n-1)(n-2)}{2}$, then X is a Hilbert space and so $\dim\big(Z(X)\big) = \frac{n(n-1)}{2}$.
> (b) $\dim\big(Z(X)\big) = \frac{(n-1)(n-2)}{2}$ if and only if X is a non-Euclidean absolute sum of $\mathbb{R}$ and a Hilbert space of dimension $n-1$.
>
> For low dimensions, Problem 27 has been solved in [74]. When the dimension of X is 3, the above result leaves only the following possible values for the dimension of $\mathcal{Z}(X)$: 0 as for $X = \ell_\infty^3$, 1 as for $\mathbb{R} \oplus_1 \mathbb{C}$, and 3 as for ℓ_2^3. When the dimension of X is 4, the possible values of the dimension of $\mathcal{Z}(X)$ allowed by the above result are $0, 1, 2, 3, 6$; all of them are possible [74, pp. 443]. The first dimension in which Problem 27 is open is $n = 5$.
>
> **Problem 28.** *What are the possible values for the dimension of $\mathcal{Z}(X)$ when X is a 5-dimensional real Banach space?*

Result

Theorem 1. *Let X be a seven-dimensional real Banach space. Then*

$$\dim \mathcal{Z}(X) \in \{0, 1, 2, 3, 4, 5, 6, 7, 9, 10, 11, 15, 21\}.$$

Conversely, every value in this set is attained by some seven-dimensional real Banach space.

This is a partial result for the general Problem 27 of the source paper: it settles the next dimension after the run's existing five- and six-dimensional packets.

Compact representation input

We use the following standard low-dimensional facts about compact Lie algebras and their orthogonal representations. Here the center acts through the skew-symmetric commutant of the semisimple part.

Lemma 1. *Let $\mathfrak{g} \subseteq \mathfrak{so}(7)$ be a compact Lie algebra, let $\mathfrak{s} = [\mathfrak{g}, \mathfrak{g}]$, and suppose $\dim \mathfrak{g} \leq 14$. Then the following table contains all cases in which $\mathfrak{s}$ can act faithfully on $\mathbb{R}^7$.*

$\mathfrak{s}$	$\dim \mathfrak{s}$	*faithful real form in dimension ≤ 7*	*max. center dim.*
0	0	*trivial*	3
$\mathfrak{su}(2)$	3	$3, 4, 5, 7$*-dimensional irreducibles, with trivials*	2
$\mathfrak{su}(2) \oplus \mathfrak{su}(2)$	6	4*-dimensional tensor form, or* $3+3$*, with trivials*	1
$\mathfrak{su}(2)^3$	9	$4+3$	0
$\mathfrak{su}(3)$	8	$\mathbb{C}^3$ *as* $\mathbb{R}^6$*, plus at most a line*	1
$\mathfrak{so}(5)$	10	$\mathbb{R}^5$*, plus at most two trivial dimensions*	1
$\mathfrak{g}_2$	14	$\mathbb{R}^7$*, the standard representation*	0

In particular, no compact Lie subalgebra of $\mathfrak{so}(7)$ has dimension 12 or 13.

Proof. A compact Lie algebra is the direct sum of its center and compact simple summands. The compact simple summands of dimension at most 14 admitting a non-trivial real representation of dimension at most 7 are

$$\mathfrak{su}(2), \qquad \mathfrak{su}(3), \qquad \mathfrak{so}(5), \qquad \mathfrak{g}_2.$$

Their relevant smallest faithful real representations have dimensions 3 or 4 for $\mathfrak{su}(2)$, 6 for $\mathfrak{su}(3)$, 5 for $\mathfrak{so}(5)$, and 7 for $\mathfrak{g}_2$.

The entries for sums of $\mathfrak{su}(2)$'s follow by the same dimension count. A faithful representation of $\mathfrak{su}(2)^2$ of dimension at most 7 is either the real four-dimensional tensor form of the two fundamental representations, with trivial summands allowed, or the sum of the two three-dimensional adjoint representations, with at most one trivial summand. A faithful representation of $\mathfrak{su}(2)^3$ in dimension at most 7 must be the four-dimensional tensor form for two factors plus the three-dimensional adjoint representation for the third. Four $\mathfrak{su}(2)$ factors already require dimension at least 8.

The center of $\mathfrak{g}$ must embed into the skew-symmetric commutant of the semisimple action. The ranks shown in the last column are immediate from the displayed decompositions: for example, a four-dimensional $\mathfrak{su}(2)^2$ tensor block has no skew commutant and the largest possible trivial complement has dimension 3, whose skew rank is 1; the $\mathfrak{su}(3)$ standard real representation has one-dimensional skew commutant generated by the complex structure; and the $\mathfrak{so}(5)$ standard representation with a trivial two-plane has at most the $\mathfrak{so}(2)$ center on that plane. These bounds also exclude dimensions 12 and 13. ∎

Lemma 2. *Let X be a finite-dimensional real Banach space and suppose $\mathcal{Z}(X)$ contains a compact semisimple algebra acting on an invariant Euclidean block in one of the following standard ways:*

$$\mathfrak{su}(3) \curvearrowright \mathbb{C}^3 \cong \mathbb{R}^6, \qquad \mathfrak{g}_2 \curvearrowright \mathbb{R}^7.$$

Then the corresponding block is actually invariant under the full orthogonal group of that block.

Proof. The connected groups SU(3) on $\mathbb{C}^3$ and G_2 on $\mathbb{R}^7$ are transitive on the Euclidean unit spheres of their standard representations. Hence a norm invariant under either group depends on the block variable only through its Euclidean length, with any remaining coordinates held fixed. Therefore the norm is invariant under the full orthogonal group of that block. □

Proof of Theorem 1

The source paper recalls two facts. First, by Auerbach's theorem one may put an inner product on X so that $\mathcal{Z}(X)$ is a compact Lie subalgebra of $\mathfrak{so}(7)$. Second, Rosenthal's gap theorem says that for $\dim X = n$:

$$\dim \mathcal{Z}(X) > \frac{(n-1)(n-2)}{2} \quad \Longrightarrow \quad X \text{ is Hilbert,}$$

so the dimension is $n(n-1)/2$; and equality at $(n-1)(n-2)/2$ occurs exactly for a non-Euclidean absolute sum of $\mathbb{R}$ and a Hilbert space of dimension $n-1$.

For $n = 7$, this leaves only dimensions at most 15, together with the Hilbert value 21. The equality case 15 is possible and already described by Rosenthal. Thus it remains to rule out the values

$$8, \quad 12, \quad 13, \quad 14.$$

Lemma 1 rules out 12 and 13 even as dimensions of compact Lie subalgebras of $\mathfrak{so}(7)$.

If $\dim \mathcal{Z}(X) = 8$, Lemma 1 forces the semisimple part to be $\mathfrak{su}(3)$ acting on a six-dimensional block $\mathbb{C}^3$, with at most one fixed real line. By Lemma 2, the norm is then invariant under SO(6) on that block, so $\mathcal{Z}(X)$ contains $\mathfrak{so}(6)$ and has dimension at least 15, a contradiction.

If $\dim \mathcal{Z}(X) = 14$, Lemma 1 forces the standard $\mathfrak{g}_2$ action on $\mathbb{R}^7$. Since G_2 is transitive on the unit sphere in $\mathbb{R}^7$, Lemma 2 implies that the norm is Euclidean and $\mathcal{Z}(X) = \mathfrak{so}(7)$, of dimension 21. This is again a contradiction.

So the only possible dimensions are

$$\{0, 1, 2, 3, 4, 5, 6, 7, 9, 10, 11, 15, 21\}.$$

It remains to realize them. Let H_k denote the k-dimensional real Hilbert space. Finite ℓ_∞-sums of Hilbert blocks realize all the listed values, because the connected isometry group is the product of the orthogonal groups on the Hilbert blocks and $\dim \mathfrak{so}(k) = k(k-1)/2$:

$\dim \mathcal{Z}(X)$	X
0	ℓ_∞^7
1	$H_2 \oplus_\infty \ell_\infty^5$
2	$H_2 \oplus_\infty H_2 \oplus_\infty \ell_\infty^3$
3	$H_3 \oplus_\infty \ell_\infty^4$
4	$H_3 \oplus_\infty H_2 \oplus_\infty \ell_\infty^2$
5	$H_3 \oplus_\infty H_2 \oplus_\infty H_2$
6	$H_4 \oplus_\infty \ell_\infty^3$
7	$H_4 \oplus_\infty H_2 \oplus_\infty \mathbb{R}$
9	$H_4 \oplus_\infty H_3$
10	$H_5 \oplus_\infty \ell_\infty^2$
11	$H_5 \oplus_\infty H_2$
15	$H_6 \oplus_\infty \mathbb{R}$
21	H_7.

This proves the classification.

Verification notes

The proof has three review points:

1. Rosenthal's gap theorem is used exactly as quoted in the source paper.
2. Lemma 1 is the finite compact-representation check. It is the main human-verification target.
3. The exclusions of 8 and 14 use only the standard sphere transitivity of SU(3) on $S^5 \subset \mathbb{C}^3$ and G_2 on $S^6 \subset \mathbb{R}^7$.

The script `code/check_seven_dimension_values.py` confirms the Euclidean-block construction values and the four missing values below the Rosenthal threshold. It is a sanity check, not a proof of the representation lemma.

Novelty check

Bounded local searches checked the run registry, solution index, attempts, and parsed source for `0605781`, `dim Z(X)`, `skew-hermitian`, and the existing five- and six-dimensional packets. The run contains no previous seven-dimensional classification.

Bounded web searches on 2026-06-13 used exact or close phrases involving:

- `dim Z(X)`, `7-dimensional`, and `Banach space`;
- `skew-hermitian`, `possible values`, and `dimension Z(X)`;
- Rosenthal's phrase `The Lie algebra of a Banach space`.

They found the source arXiv page and adjacent background, but no direct later classification of the seven-dimensional case.

Scope

This packet settles only the seven-dimensional subcase of Problem 27. It does not claim the general all-dimensional classification.